

Table 1: Information provided by the different representations of an audio signal

Representation	Details available	Details missing
Time domain waveform	Amplitude, time	Frequency
Frequency spectrum	Frequency, Amplitude	Time
Spectrogram	Amplitude, frequency, time	--

Table 2: Application-specific dedicated open source toolkits for implementing various audio processing tasks

Task	Toolkits
Automatic speech recognition (ASR)	Hidden Markov model (HMM) based toolkit (HTK)[1], Kaldi[2], CMU Sphinx [3]
Speaker diarization	DiarTK, LIUM
Automatic speaker recognition (ASR) and verification (ASV)	Bob, ALIZE, SIDEKIT
Music information retrieval (MIR)	Essentia, SMIRK
Common tools for audio data analysis	Audacity, Praat

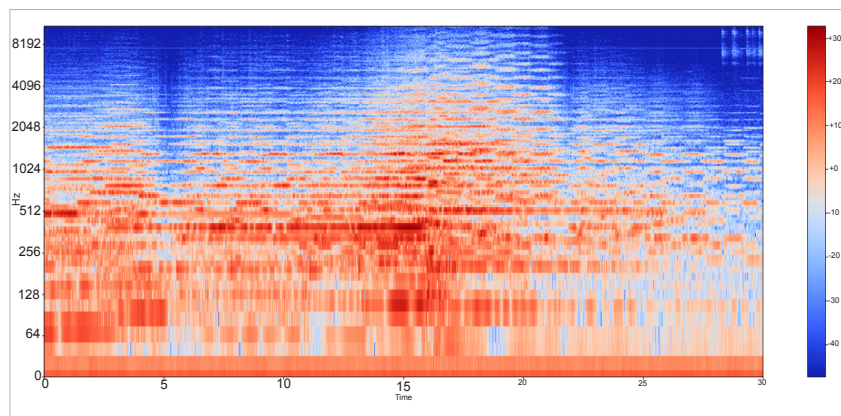


Figure 8: Spectrogram representation of the audio signal

```
def plot_spectrogram(Y, sr, hop_length,
    y_axis="linear"):
    plt.figure(figsize=(25, 10))
    librosa.display.specshow(Y, sr=sr, hop_
length=hop_length, X_axis="time", y_
axis=y_axis)
    plt.colorbar(format="%+2.f")
```

Often, signals are transformed to the frequency domain to reveal the majority of their inherent information. Time domain analysis has certain limitations, which the frequency domain can counter. Both spectrum and spectrograms are frequency domain representations of digital signals. However, in addition to time and frequency information, the latter also provide information about signal

amplitude (loudness). Spectrogram based analysis is sometimes referred to as time-frequency analysis.

Voice spectrography analysis is used in crime investigations to match a suspected speaker's voice pattern with that in criminal records. It also helps in applications such as vowel recognition based on frequency components of the signal.

Recent trends

Due to the pandemic, everyone prefers touch-free and hands-free interfaces for interacting with applications. Voice based devices are therefore being used for various applications. Audio spoofing detection is a recent technology, where the system checks whether the voice is generated from a genuine user or is a fake one. An interesting area of research in health care is analysing audios to determine the symptoms of any respiratory disease. In the field of music, speech-music discrimination can be done by removing or subduing the speech parts of the signal, and emphasising only the music.

Now that you have read this article, I am sure you are eager to experiment with audio signals. To begin that journey, just record your voice and play with the code given in this article. You may be amazed at what you discover along the way. **END** 🐼

References

- [1] <https://cmusphinx.github.io/wiki/tutorial/>
- [2] <http://kaldi-asr.org/doc/about.html>
- [3] https://www.speech.kth.se/~matsb/speech_rec_course_2003/htk_tutorial.pdf
- [4] <https://projets-lium.univ-lemans.fr/spkdiarization/>
- [5] <https://alize.univ-avignon.fr/>
- [6] <https://www.idiap.ch/software/bob/>
- [7] <https://projets-lium.univ-lemans.fr/sidekit/>
- [8] <https://essentia.upf.edu/>
- [9] <https://smirk.cs.princeton.edu/>
- [10] Garg S., Mankad S.H., Voice liveness detection under feature fusion and cross-environment scenario, *Multimed Tools Appl* 79, 26951–26967 (2020)

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